

SOLUTIONS

Name: _____ Date: _____



Year 10 (Pre stage 2) In Class Investigation Transforming Quadratics

Total Time: 55 minutes

Weighting: _____ % of the year.

Equipment: CAS calculator

23	%
=	%

Penalties (only deducted once in the whole assessment):

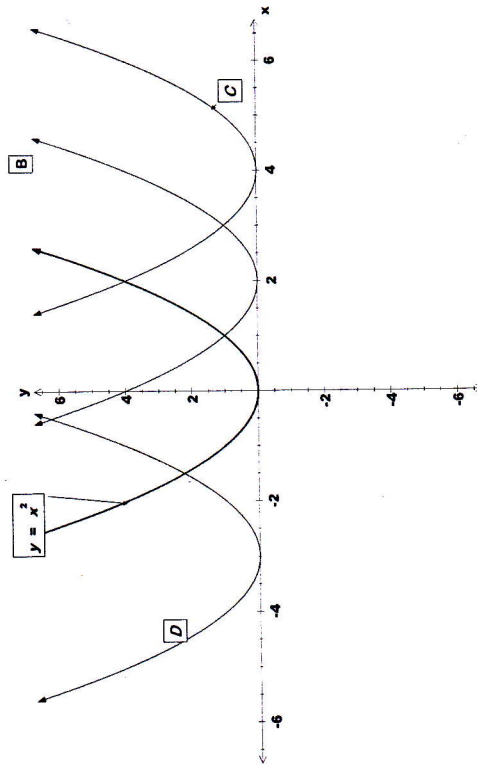
- ☐ Rounding - Incorrect/inappropriate (-1)
- ☐ Notation - incorrect/missing (-1)
- ☐ Units - Incorrect/missing (-1)

Important Information:

This investigation will be completed under test conditions. Answers should be rounded appropriately. All working should be shown in the space provided. Solutions without working may not be awarded full marks. Please take the marks for each question into account when answering the question.

1. [3 marks: 1, 1, 1]

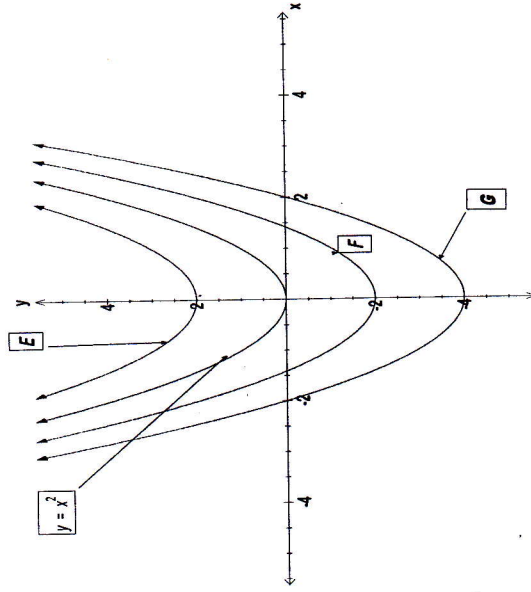
In the graph below the curves B, C and D are all of the same shape as the curve A. If the curve A has equation $y = x^2$. Write down the equations of curves B, C and D.



- i) Equation of B $y = (x-2)^2$ ✓
- ii) Equation of C $y = (x-4)^2$ ✓
- iii) Equation of D $y = (x+3)^2$ ✓

2. [9 marks: 1,1,1,2,2,2]

In the graphs below the curves E, F, G, H, I and J are all of the same shape as the $y = x^2$ curve. Write down the equations of the curves E, F, G, H, I and J.



Equation of curve E

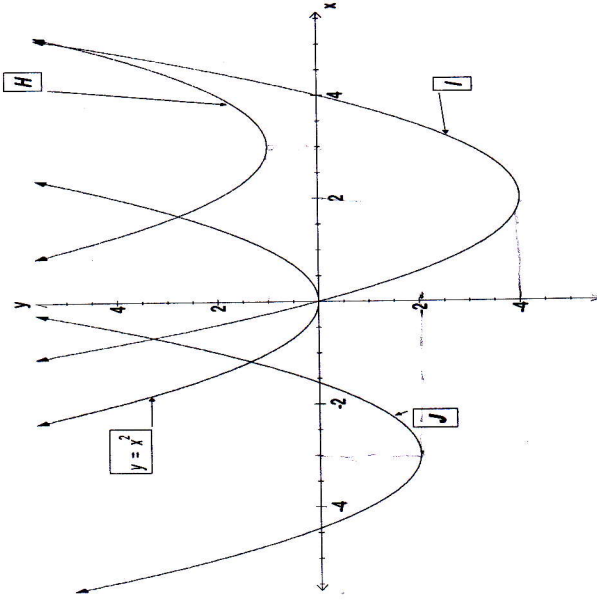
$$y = x^2 + 2 \quad \checkmark$$

Equation of curve F

$$y = x^2 - 2 \quad \checkmark$$

Equation of curve G

$$y = x^2 - 4 \quad \checkmark$$



Equation of curve H

$$y = (x-3)^2 + 1 \quad \checkmark$$

Equation of curve I

$$y = (x+2)^2 - 4 \quad \checkmark$$

Equation of curve J

$$y = (x+3)^2 - 2 \quad \checkmark$$